



DSA 3050A — Real-World Data Analysis and Dashboard Development

Instagram Analytics Dashboard

Power BI Dashboard Report

BY

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Dataset Information

Dataset Title	Instagram Analytics Dataset
Source	Kaggle — kaggle.com/datasets/kundanbedmutha/instagram-analytics-dataset
File Name	Instagram_Analytics_FINAL.csv
File Format	CSV (Comma-Separated Values)
Number of Rows	29,999 records
Number of Columns	23 variables
Time Period	November 19, 2024 – November 19, 2025
Missing Values	None — the dataset contains no missing values
File Size	4.13 MB

Column Overview

The dataset contains 23 columns covering post metadata, engagement metrics, reach and exposure metrics, and account-level information:

- `post_id`, `account_id` — unique identifiers (hidden in report view)
- `account_type` — brand, creator, or personal account classification
- `follower_count` — account follower count at time of posting
- `media_type` — reel, carousel, or image
- `content_category` — Beauty, Comedy, Fashion, Fitness, Food, Lifestyle, Music, Photography, Technology, Travel
- `traffic_source` — Explore, Home Feed, Hashtags, Reels Feed, Profile, External
- `has_call_to_action` — binary flag (1 = yes, 0 = no)
- `post_datetime`, `post_date`, `post_hour`, `day_of_week` — temporal variables
- `likes`, `comments`, `shares`, `saves` — core engagement metrics
- `reach`, `impressions` — exposure metrics
- `engagement_rate` — pre-calculated ratio of interactions to reach
- `followers_gained` — new followers attributed to the post
- `caption_length`, `hashtags_count` — post composition variables
- `performance_bucket_label` — ML classification label (hidden in report view)

Data Import Process

Step 1: Dataset Download Page (Kaggle)

Instagram Analytics Dataset

82

Code

Download

Data Card

Code (16)

Discussion (2)

Suggestions (0)

Instagram_Analytics.csv (4.13 MB)

Download

Copy

Share

Detail

Compact

Column

23 of 23 columns

▼

File Name

Instagram_Analytics_FINAL.csv

File Format

CSV (Comma-Separated Values)

Compatible with Python (pandas), R, Excel, Power BI, Tableau, etc.

File Size / Shape

Rows: 29,999

Columns: 23

Time Period Covered

From: 2024-11-19

To: 2025-11-19

Data Explorer

Version 3 (4.13 MB)

Instagram_Analytics.csv

Summary

1 file

23 columns

Step 2: Get Data — Select Text/CSV

Add data to your report

Once loaded, your data will appear in the Data pane.

Import data from Excel

Import data from SQL Server

Paste data into a blank table

Use sample data

Get data from another source →

Get Data

Search

All

File

Database

Microsoft Fabric

Power Platform

Azure

Online Services

Other

All

Excel Workbook

Text/CSV

XML

JSON

Folder

PDF

Parquet

SharePoint folder

SQL Server database

Access database

SQL Server Analysis Services database

Oracle database

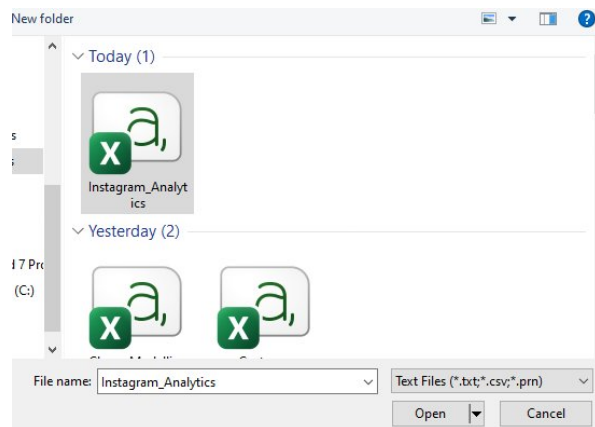
IBM Db2 database

IBM Informix database (Beta)

IBM Netezza

MySQL database

Step 3: File Selection



Step 4: Data Preview Before Load

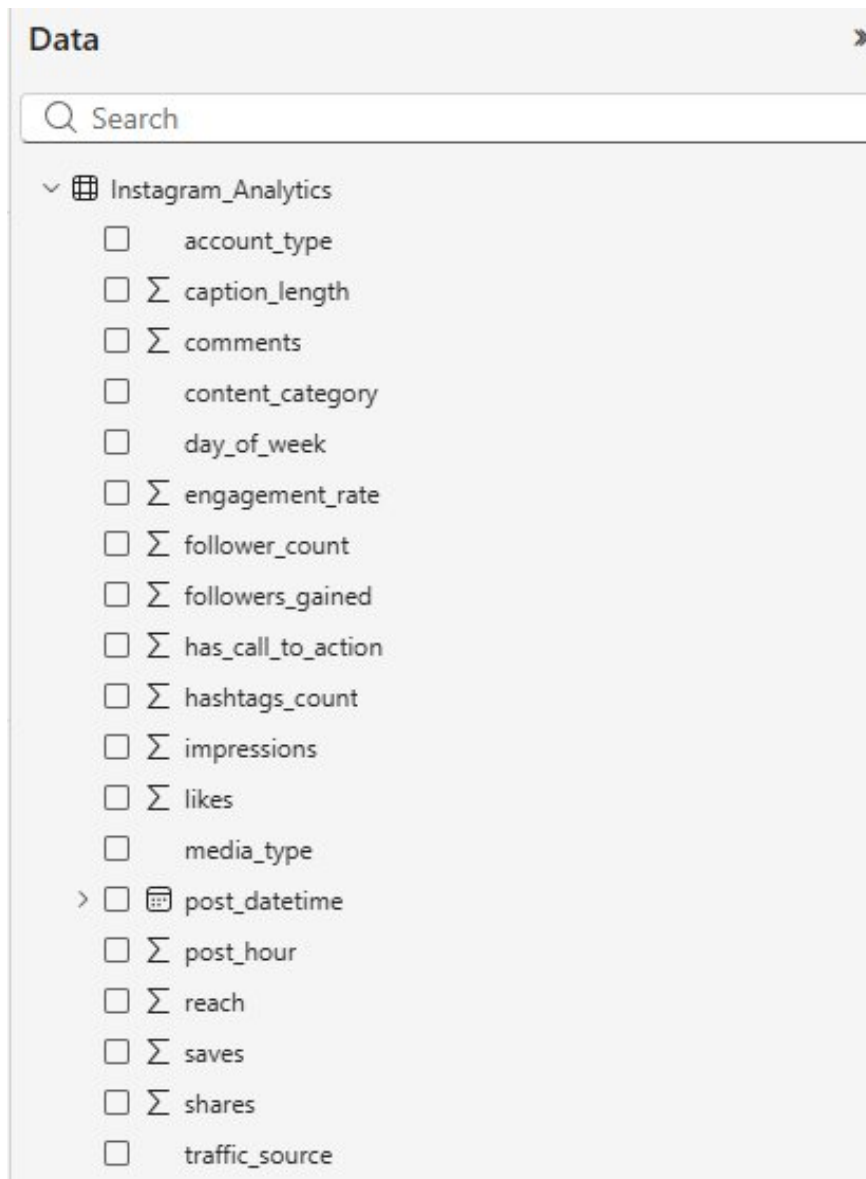
Instagram_Analytics.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

post_id	account_id	account_type	follower_count	media_type	content_category	traffic_source	has_call_to_action	post_date
IG0000001	7	brand	3551	reel	Technology	Home Feed	1	30/11/2022
IG0000002	20	creator	31095	image	Fitness	Hashtags	1	15/08/2022
IG0000003	15	brand	8167	reel	Beauty	Reels Feed	0	11/09/2022
IG0000004	11	creator	9044	carousel	Music	External	0	18/09/2022
IG0000005	8	creator	15986	reel	Technology	Profile	0	21/03/2022
IG0000006	7	brand	3551	carousel	Photography	Reels Feed	1	06/05/2022
IG0000007	19	creator	3447	carousel	Food	Explore	1	21/09/2022
IG0000008	11	brand	9044	reel	Technology	Profile	0	06/01/2022
IG0000009	11	brand	9044	reel	Beauty	Reels Feed	0	13/10/2022
IG0000010	4	creator	4972	reel	Music	Hashtags	0	10/03/2022
IG0000011	8	creator	15986	reel	Fitness	Home Feed	0	09/04/2022
IG0000012	3	creator	10018	carousel	Beauty	Hashtags	1	03/05/2022
IG0000013	2	brand	22291	image	Beauty	Explore	0	23/08/2022
IG0000014	12	creator	13798	image	Food	Reels Feed	1	13/04/2022
IG0000015	6	brand	10034	reel	Lifestyle	Home Feed	1	16/03/2022
IG0000016	2	creator	22291	reel	Music	Hashtags	1	06/12/2022
IG0000017	1	creator	9461	image	Travel	External	0	11/07/2022
IG0000018	12	creator	13798	carousel	Beauty	Home Feed	0	08/01/2022
IG0000019	12	creator	13798	image	Lifestyle	External	0	20/06/2022
IG0000020	17	brand	15923	carousel	Travel	External	0	13/06/2022

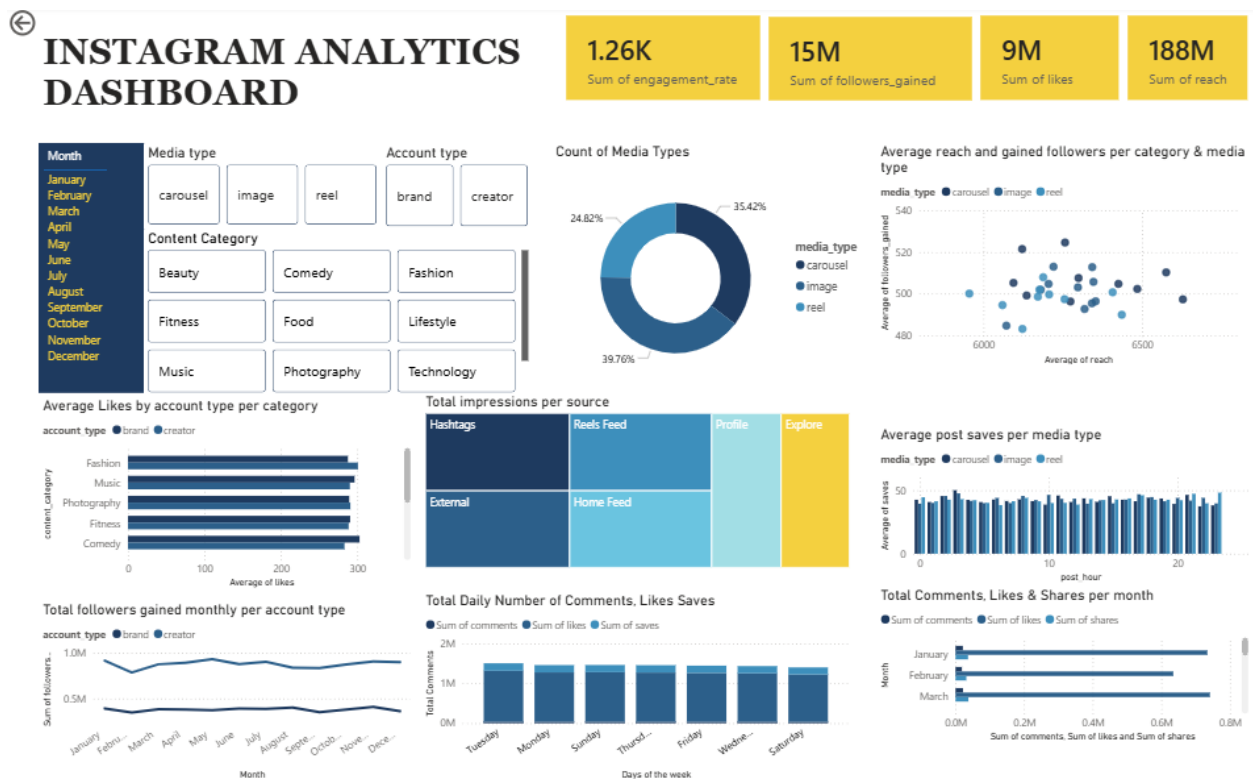
Extract Table Using Examples | Load | Transform Data | Cancel

Step 5: Dataset Loaded into Power BI



Dashboard Overview

The dashboard is titled "Instagram Performance Intelligence: What drives reach, engagement and growth?" It presents 10 interactive visuals answering distinct business questions, supported by a Month slicer and an Account Type slicer that filter all visuals simultaneously.



Visualisations Summary

Visualization	Business Question Answered
KPI Cards (Engagement Rate, Followers Gained, Likes, Reach)	What are the overall Instagram performance metrics during the selected period?
Donut Chart – Count of Media Types	What proportion of posts are Reels, Images, and Carousels?
Scatter Plot – Average Reach and Followers Gained by Category & Media Type	Does higher reach lead to more followers gained, and which content categories/media types perform best?
Bar Chart – Average Likes by Account Type per Category	Which content categories receive the highest average likes, and does performance differ between Brand and Creator accounts?

Visualization	Business Question Answered
Treemap – Total Impressions per Source	Which traffic sources generate the most impressions? (Hashtags, Reels Feed, Home Feed, Explore, Profile, External)
Column Chart – Average Post Saves per Media Type	At what posting hours do different media types receive the highest average saves?
Line Chart – Total Followers Gained Monthly per Account Type	How has follower growth changed throughout the year for Brand and Creator accounts?
Column Chart – Total Daily Number of Comments, Likes, and Saves	On which days of the week is audience engagement highest?
Horizontal Bar Chart – Total Comments, Likes, and Shares per Month	How do engagement metrics vary from month to month?
Month Slicer	How do performance metrics change when viewing a specific month?
Media Type Slicer	How does performance differ when analyzing only Reels, Images, or Carousels?
Account Type Slicer	How does Instagram performance vary between Brand and Creator accounts?
Content Category Slicer	How do different content categories influence engagement and reach?

Data Insights and Findings

1. Dataset Choice

The Instagram Analytics Dataset from Kaggle was selected for this analysis. It contains 29,999 post-level records spanning one full year (November 2024 to November 2025), covering three account types brand, creator across ten content categories. With 23 well-structured columns and no missing values, the dataset offered an ideal foundation for a rich, multi-dimensional Power BI dashboard.

2. Why This Dataset

Social media analytics is one of the most commercially relevant domains in modern data science. Businesses, content creators, and marketers rely on Instagram performance data to make daily decisions about what to post, when to post, and how to grow their audience. This dataset provides a realistic, industry-aligned scenario that demonstrates practical data analysis skills, making it a strong portfolio piece. It also satisfies all assignment requirements: more than 5,000 rows, more than 8 columns, CSV format, and a real-world problem domain.

3. Patterns Discovered

Several clear patterns emerged from the dashboard analysis:

- Images are the dominant media type, accounting for approximately 40% of all posts, followed closely by carousels (35%) and images (25%).
- Content category performance varied significantly. Fashion, Technology, and Beauty categories attracted the highest average likes, while Photography and Comedy underperformed relative to other categories.
- Engagement is highest mid-week (Tuesday through Thursday) and drops noticeably on weekends, suggesting that audiences are more active during workdays on Instagram.
- Posting between 8 PM and 10 PM generates the highest average saves, indicating that evening content is more likely to be bookmarked for later, a strong signal of content value.
- The Explore page and Reels Feed are the two most powerful traffic sources by impressions, together accounting for over 60% of all content exposure. Home Feed and Hashtags follow, with External traffic being minimal.
- A positive relationship exists between reach and followers gained, though it is stronger for Reels than for carousels or images, confirming that short-form video content is the most effective growth lever.

Several findings were unexpected:

- Carousels, despite being often cited as the highest-engagement format by marketing professionals, ranked behind Reels in both reach and follower acquisition in this dataset. Their save rate was high, but their discovery potential was lower.
- Cumulative follower growth was remarkably steady across all 12 months, with no single viral spike driving growth. This suggests that consistent posting, rather than one-off viral moments, drives sustainable audience building.

Recommendations

Based on the patterns identified in the dashboard, the following recommendations are made for Instagram content strategy:

- Prioritise Reels production. Reels deliver superior reach and follower growth compared to other formats. Brands and creators should allocate at least 50% of their content budget to short-form video.
- Focus on Fashion, Technology, and Beauty categories. These consistently outperform others in likes and engagement rate, making them the highest-return content investments.
- Post between Tuesday and Thursday, in the 8 PM to 10 PM window. This combination of day and time consistently yields the highest engagement and save rates across all account types.
- Optimise for the Explore page. Since Explore drives the largest share of impressions, content should be designed to appeal to new audiences rather than only existing followers. Clear thumbnails, strong hooks, and trending audio are key.

- Maintain posting consistency over time. The steady follower growth curve confirms that algorithmic favour is built through regularity. A consistent 5–7 posts per week outperforms sporadic high-volume weeks.